

Fall 2025.

Physics 509. Quantum Theory I.
Yurii Shylnov.

Homework # 3.

Due to October 8th, 2025.

You can use any source to solve the problems. However, your solutions should be presented in a very detailed and logical way. You need to understand every step of your solution and be able to explain it.

1. Prove that if $[\hat{A}_S, \hat{B}_S] = \hat{C}_S$, where $\hat{A}_S$, $\hat{B}_S$, and $\hat{C}_S$ are operators in the Schrödinger representation, then an equal time commutation relation for the operators in the Heisenberg representation, $\hat{A}_H(t)$, and $\hat{B}_H(t)$, is the same, $[\hat{A}_H(t), \hat{B}_H(t)] = \hat{C}_H(t)$.
2. For a one-dimensional harmonic oscillator find an explicit form for the operators of a linear momentum and coordinate in the Heisenberg representation using a) the Heisenberg evolution equation; b) the original formula for a transition from the Schrödinger to the Heisenberg representation of operators and the Baker-Hausdorff lemma.
3. For a one-dimensional harmonic oscillator find an explicit form for the operators of creation and annihilation in the Heisenberg representation using a) the Heisenberg evolution equation; b) the original formula for a transition from the Schrödinger to the Heisenberg representation of operators and the Baker-Hausdorff lemma.
4. For a one-dimensional harmonic oscillator, what are the equal-time commutation relations in the Heisenberg representation of a) the operators of a linear momentum and coordinate; b) the creation and annihilation operators?
5. For a one-dimensional harmonic oscillator calculate the non-equal time commutation relations in the Heisenberg representation of a) the operators of a linear momentum and coordinate, $[\hat{p}_H(t), \hat{x}_H(t')]$; b) the creation and annihilation operators, $[a_H(t), a_H^\dagger(t')]$.